

GBI Genome Atlas: how to change it

The spreadsheet in Google Drive is the single source. Edit it there and the web page rebuilds from it. Nothing is stored in two places, and you never need to open code.

TO CHANGE ANYTHING

1. Open the master spreadsheet in Drive. Edit the cell.
2. In GitHub, go to **Actions**, choose **Build and publish atlas**, press **Run workflow**.
3. About a minute later the live page shows your change. A green tick means it worked.

A failed build cannot break the live page. It keeps serving the last version that worked, and emails whoever ran it with the reason. Nothing broken is ever public.

WHAT YOU CAN EDIT

- **RESULTS** All 218 plant records: names, rarity rank, accession numbers and their links, collecting institution, bloom months, genome size, N50, gene counts, and which pipeline stages each plant reached. Adding an accession as GenBank issues it is a cell edit.
- **BUSCO GENE LIST v2**, **BARCODE PRIMER LIST v2**, **TF BY FAMILY v2** The three matrix pages.
- **ATLAS Copy** **All 209 pieces of writing on the site**, grouped by page in the **tab** column: the opening paragraph on the home page, every page heading and standfirst, the whole Method and gaps page, all chart and plate captions, the collection and provenance notes, the plastome plate notes, the BUSCO and barcode captions, the rarity rank descriptions, the enzyme family descriptions, and the status messages inside the plant drawer. Edit the **text** column. Never change the **key** column.
- **ATLAS Comparisons** Every published figure the atlas sets itself against, with its source, link and the date it was checked. Each figure is written once here and used everywhere it appears, so the page cannot contradict itself.
- **ATLAS Regulators** The 30 reference gene-family counts.
- **ATLAS Settings** Contact email, number of embargoed plants and their release date, image licence.

TEXT WITH {BRACES} IN IT

23 of the 209 rows on **ATLAS Copy** contain placeholders such as `{n}`, `{max}` or `{pct}`. The build replaces each one with a figure counted from the data at the moment the page is made, which is why those sentences never go stale. Rewrite the wording around them however you like, and move them within the sentence, but keep the braces and the exact spelling inside them or the number will not appear.

IF THE PAGE LOOKS WRONG

Work through these in order. It is almost always the first one.

- **Did the build actually run?** Check the Actions tab. A red cross means it stopped, and the message names the cell or key at fault. The live page is still the previous good version.
- **Is a word or heading missing entirely?** Look for an unclosed `<b>` or `<a>` in the text cell you last edited. Broken formatting is the one mistake the build cannot see.
- **Did a number turn into literal text like {n}?** The placeholder was renamed or a brace was lost. Restore the exact spelling from an untouched row.
- **Is it only the photographs?** That is Calflora, not the atlas. Check your connection first, then whether Calflora itself is reachable.

WHAT YOU CANNOT CHANGE FROM THE SPREADSHEET

- **The Chemistry, Markers and Plastomes findings.** The numbers on those three pages come from finished analysis files, not from the spreadsheet. The tabs **CHEMISTRY v2**, **SSR PANEL v2** and **CHLOROPLAST v2** are *not* read by the atlas. Editing them changes nothing. The writing on those pages is on **ATLAS Copy** like everything else.
- **Any figure counted from the plants.** The tiles across the top of the home page, the rarity breakdown, the pipeline funnel, the counts beside each tab. These are recounted from the 218 records every time the page is built, so the summary can never disagree with the data underneath it. To change them, change the plants.
- **Nine chart captions** that quote their own chart's numbers back at you, on Barcodes, Bloom, Chemistry, Landscape, Regulators and in the plant drawer. They are written in code so the figure in the sentence cannot drift from the bar above it.
- **Layout, colour, chart types, anything that moves or responds.** Also adding a new page, and the 182 chloroplast plate images. Those are code and files.

MISTAKES THE BUILD WILL CATCH

In each case the build stops, names the problem, and the live page is untouched.

- Deleting a row from **ATLAS Copy** that the site uses. It names the key.
- Adding a row whose key nothing reads, so you are not left editing a cell that does nothing.
- Inserting or deleting a column on **RESULTS**. You get **COLUMN SHIFT at index N**. This has gone wrong once before, which is why the check exists.
- Renaming or deleting one of the tabs above. Capitals and underscores do not matter; the words do.
- A build that somehow produces no chloroplast plates. It refuses to publish rather than quietly ship a page with the maps missing.

Safe, and tested: sorting or reordering rows on any **ATLAS** tab, and adding your own notes in columns to the right. Rows are found by name, not by position.

TWO MISTAKES IT WILL NOT CATCH

- **Broken formatting inside a text cell.** Copy may contain simple HTML such as `<b>`, `<em>` and links. An unclosed tag will render wrong without failing the build. If a page looks mangled after an edit, check this first.
- **Deleting a plant row.** The plant simply vanishes from the atlas. That is intended, but it happens silently, so be deliberate.

THREE RULES

- **Do not move the spreadsheet or restrict who can view it.** The build reads it from Drive at that address. If it moves or is locked down, every rebuild fails until it is restored.
- **The GitHub repository must stay public.** Free hosting depends on it.
- **Photographs are not stored with the site.** They load from Calflora under CC BY-NC 4.0 with each photographer credited. If Calflora ever changes its image addresses, the photographs stop appearing, and no change here can prevent that.